

Fig. 6: AC mains rectifier with inrush current limiter and high voltage reservoir



Fig. 7: Capacitor discharger probe with test leads and LED indication

For example, a capacitor charged to 600V DC and discharged through this circuit produces an initial current of approximately 54mA, calculated as $(600V - 5.6V)/11k\Omega$. Although instantaneous power dissipation in the bleeder resistors briefly exceeds their combined 10W rating, this condition lasts only a short time as the voltage decays rapidly. Peak Zener dissipation is about 300mW ($5.6V \times 54mA$), which remains well within its rating.

During practical testing, a 150μF/450V capacitor was charged to approximately 320V DC using the circuit shown in Fig. 6. To discharge it, the mains power was removed and the circuit shown in Fig. 5 was used. The discharge time was approximately 10-12 seconds, depending on the conditions.

Fig. 7 shows the completed discharger probe system with insulated test leads and LED indicators, designed specifically for the safe handling of high-voltage capacitors.

EFY warning! Capacitors can retain charge for long periods, and in some systems, the stored energy can be dangerous or even lethal. Although the system appears straightforward, it involves working with hazardous high voltages. Construction or use should be attempted only by individuals familiar with safe high-voltage handling practices. **EFY**



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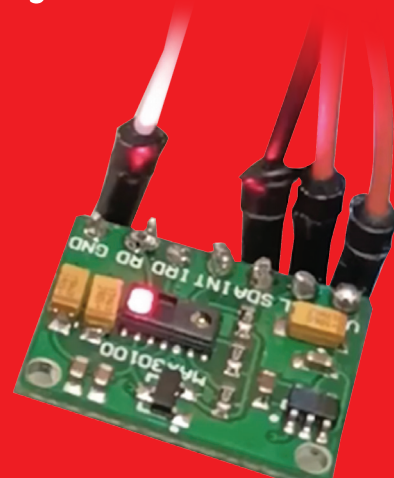
This indicator network is connected in parallel with a 5.6V/1W Zener diode (ZD1), which limits the voltage across the LEDs and protects them during high-voltage discharge.

When the probe is connected to a charged capacitor, the LEDs illuminate and gradually dim as the voltage decreases. The capacitor's stored energy powers the LEDs, offering a clear visual indication of the discharge process and helping determine when the voltage has fallen to a safer level.

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